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Survey paper

Traffic management approaches using machine learning and deep learning techniques: A survey

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ABSTRACT

Traffic management is improved in cutting-edge smart cities using technologies such as machine learning and deep learning to streamline daily tasks and boost productivity. However, traffic management still suffers from challenging issues such as poor traffic congestion prediction, lack of traffic flow management, public transportation optimization, and emergency management. In this article, we provide a thorough understanding of the benefits, drawbacks, and practical implications of leveraging machine learning and deep learning in Traffic Management Systems (TMSs) by methodically reviewing and critically analysing various traffic management techniques. We also present a generic traffic management architecture that uses a set of assessment criteria to evaluate 20 recently proposed research prototypes (i.e., published since 2019). Finally, we highlight the ongoing challenges and prospective trajectories within the rapidly evolving domain of traffic management, underscoring the need to address emerging issues and directions in its dynamic development. This survey article offers insights that might help in efficiently tackling the issues posed by traffic management while maximizing the potential of machine learning and deep learning techniques. This survey can be a significant resource for researchers, policymakers, and practitioners.

1. Introduction

The orchestration of automotive and pedestrian mobility over public thoroughfares, including roads, streets, and highways, is referred to as "traffic management". At its core, traffic management aims to promote the safe and effective circulation of traffic, relieve congestion, and lower the number of collisions on the roads. These goals require the use of a wide range of strategies, including traffic engineering, the use of traffic control equipment, and careful transportation planning. All of these actions work to regulate vehicle movement and ensure the safety of all road users (Nellore and Hancke, 2016).

In modern civilizations, traffic management is crucial, especially in urban areas where the pervasive problem of traffic congestion looms large. Significant economic consequences of this congestion include lost productivity, wasted fuel, and increased air pollution. By improving traffic flow and reducing the amount of time commuters spend stuck in traffic, effective traffic management can lessen these effects, reducing stress and improving the well-being of both drivers and passengers (Alsrehin et al., 2019). People in general and tourists alike struggle with the effects of increased traffic and interruptions brought on by population development. Based on elements like accidents, road work, special events, and unlawful parking, it is possible to distinguish between regular and non-recurring traffic obstacles. Accidents stand out among them due to their direct effect on human life and their disproportionately high rate of fatalities among children.

Our society is experiencing the integration of intelligent technologies that improve our daily activities in an era of developing technology. A significant technical development known as the Internet of Things (IoT) connects many smart gadgets, allowing for smooth connection and data exchange. IoT technology is now more widely used in the transport sector, where it has a significant impact on how effectively transport systems operate. To further improve these systems, information technology must be integrated more. As travel demands continue to diversify, traditional traffic management methods are no longer adequate. Governments, such as China, have recognized the need for scientific and advanced traffic management approaches, which necessitate the integration of emerging Artificial Intelligence (AI) technologies (Cheng et al., 2021). To detect and prevent traffic-related concerns, researchers have resorted to Machine Learning (ML), Deep Learning (DL) techniques, and the quantity of geo-tagged data available on social media platforms (Azhar et al., 2023; Abdullah et al., 2023) which improved traffic management strategies.

Congestion prediction is given top priority by effective traffic management centres globally to provide a quick incident reaction. Due to the interdependencies of traffic flow in both time and space, this work is

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difficult. The ever-growing amounts of traffic data generated have been used to train ML predictions, but DL approaches have shown promise as alternatives and outperformed conventional models. Nevertheless, there are still questions about their applicability, accuracy, and ideal parameter adjustment (Mihaita et al., 2020). In fact, as traffic control centres work to maintain regular travel patterns and assure prompt incident clearing on a daily basis, traffic congestion continues to be a major concern (Jovanović et al., 2022). The use of deep learning in IoT applications is still in its infancy despite substantial efforts in recent years (Ma et al., 2019).

The research questions we aim to address are as follows:

- What machine learning and deep learning techniques have been used in traffic management systems?
- What are the advantages, disadvantages, and practical implications of using machine learning and deep learning in traffic management systems?
- How can a generic traffic management architecture be evaluated using predefined assessment criteria to handle traffic management challenges?

In this work, we overview the benefits, drawbacks, and practical implications of leveraging ML and DL in TMSs by methodically reviewing and critically analysing various traffic management techniques. In addition, we present a generic traffic management architecture that uses 12 different assessment criteria among 3 layers to evaluate 20 recently proposed research prototypes. Furthermore, we highlight current research issues and future research directions in the field of traffic management. This survey article is considered a guideline for researchers and practitioners who are planning to leverage ML and DL techniques to enhance their traffic management systems.

The rest of this article is organized as follows. Section 2 overview the background. Related Work is discussed in Section 3. Section 4 illustrates the design of the traffic management framework architecture. Section 5 presents the research prototype evaluation. Section 6 provides the traffic management open issues and future directions. Finally, concluding remarks are presented in Section 7.

2. Background

We give a brief overview of traffic management in this section. The core ideas and guiding principles of machine and deep learning are then introduced. We also go through popular deep-learning models that are useful for analysing, detecting, and predicting traffic data.

2.1. Overview of traffic management

Metropolitan regions around the world are plagued by traffic, which has a serious impact on quality of life, economic productivity, and urban mobility. The number of vehicles on the road has increased along with city growth and population growth, causing traffic that is detrimental to travel efficiency and the performance of the entire transportation system. Queues, slower speeds, longer travel times, and a general degradation in service quality on road networks are the hallmarks of traffic (Cvetek et al., 2021).

Beyond just creating a minor nuisance for commuters, traffic has far-reaching effects. Roadway traffic results in fuel waste, air pollution, and greenhouse gas emissions, which have a negative impact on the environment and public health (Afrin and Yodo, 2020). Additionally, traffic jams impose significant economic consequences like increased car repair costs, wasted time, and fuel use. It is also reducing the availability of goods and services, messes up supply networks, and makes cities less competitive overall (Levy et al., 2010).

Utilizing effective traffic management strategies is crucial for improving urban transportation networks and ensuring their effectiveness. This is because traffic control measures have broad-reaching effects (Retallack and Ostendorf, 2019; Liu et al., 2023). Researchers, decision-makers, and transportation specialists have developed and implemented a variety of solutions over time with the goal of improving traffic management and its overall efficacy.

The field of traffic management includes a variety of methods and solutions designed to ease traffic congestion, improve traffic flow, and boost the efficiency of the entire transportation system. These tactics include both demand-management initiatives like improving public transportation, encouraging carpooling, and charging for congestion as well as infrastructure-based actions like road widening, intersection upgrades, and traffic signal optimization (Oladimeji et al., 2023).

Recent technological developments and the spread of data-driven methodologies have created new possibilities for traffic management. Innovative technologies, like real-time traffic monitoring, data analytics, and communication networks, are used by Intelligent Transportation Networks (ITNs) to enable more effective traffic management. Furthermore, the development of Machine Learning (ML) and Deep Learning (DL) methodologies has opened up fresh possibilities for traffic forecasting, congestion monitoring, and adaptive traffic management (Auer et al., 2016).

2.2. Machine learning

The roots of machine learning can be found in several scientific fields, including computer science, mathematics, and statistics (Sarker, 2021b). The first attempts to create machines that could learn from data were made in the 1940s, which is when machine learning began to take off. The invention of the perceptron, a class of Artificial Neural Networks (ANNs) that could learn from examples, was the first significant advancement in this field (Patil, 2019). Researchers created training techniques for perceptrons in the late 1950s and early 1960s, which paved the way for the first machine-learning application. The paucity of computer resources and data caused a slowdown in machine learning research in the 1970s and 1980s, however, this initial excitement was short-lived (Akram et al., 2023).

The introduction of the World Wide Web (WWW), which generated enormous amounts of data and offered a platform for distributed computing, was responsible for the revival of machine learning research in the 1990s (Sarker, 2021a). Support Vector Machines (SVM) and Decision Trees (DT) are examples of new machine learning algorithms that researchers created that are better suited for large-scale data processing (Sarker, 2021b). Machine learning algorithms' accuracy and robustness were greatly enhanced with the invention of ensemble learning methods like random forests and boosting. Deep learning, which includes training ANNs with several layers of neurons, first gained popularity in the early 2000s. Machine learning research in areas like speech and image identification, robotics, and Natural Language Processing (NLP) has greatly advanced thanks to deep learning (Möller, 2023).

In today's world, machine learning is a pervasive technology that is employed in a variety of industries, including marketing, banking, healthcare, and entertainment. Researchers have created increasingly complex machine learning algorithms that can learn from a variety of data sources, including social media, sensors, and electronic health records, thanks to the broad availability of data and computer capacity (Tsaramirsis et al., 2022). Research in robotics and Artificial Intelligence (AI) now has more options because of the emergence of Reinforcement Learning (RL), which includes teaching machines to learn through feedback. It is anticipated that academics will create new methods and algorithms that will expand the capabilities of machines as machine learning continues to advance (Nellore and Hancke, 2016).

For forecasting and allocating traffic flow in extensive metropolitan networks, machine learning techniques have grown in value (Medina-Salgado et al., 2022). Compared to more conventional methods of traffic assignment, these techniques have several advantages, such as the capacity to process enormous amounts of data quickly and accurately, the ability to spot patterns and trends that human observers might miss, and the flexibility to adjust to shifting traffic conditions (Miglani and Kumar, 2019). Transportation planners and engineers will have access to more advanced tools for controlling traffic flow across extensive urban networks as machine learning algorithms continue to advance (Iyer, 2021).

2.3. Deep learning

Deep Learning (DL) has grown in popularity across a wide range of areas, owing mostly to the efficacy of Artificial Neural Networks (ANNs). ANNs have evolved as a prominent data representation approach that is intended to emulate the operation of the human brain. They are made up of interconnected nodes, each with weighted connections to nodes in different layers. These nodes receive input from connected nodes and use weights and a basic function to compute output values. For a specific node, its inputs denoted ι and the weight (π) where the weighted input is summed and the result runs through a nonlinear activation function denoted γ, where z is the number of inputs for the node. This can be expressed as a vector dot product as the following equation:

$$\mu = \gamma(\iota \cdot \pi) = \left(\sum_{i=1}^{z} \iota_i \pi_i \right) \quad where \quad \iota \in d_{1\times z},\ \pi \in d_{z\times 1},\ \mu \in d_{1\times 1} \tag{1}$$

Deep Neural Networks (DNNs), in particular, have gained popularity due to their flexibility, capacity to incorporate non-linearity, and data-centric model creation (Pinto Neto et al., 2023; Noor et al., 2023). A DNN classifier normally associates a collection of classes $(\lambda_k)k$ with data points y. Encoding the modelled class probabilities $\alpha(\lambda_k|y)$ are the output neurons. We can locate a prototype y that is typical of the class λ_k by Eq. (2):

$$\max_x \ \log \ \lambda_k \ \ \alpha(\lambda_k|y) - \delta \, \|y\|^2 \tag{2}$$

Fully-Connected Neural Networks (FNNs), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs) are the three primary forms of neural networks. CNNs excel at visual tasks because they take advantage of the inherent spatial features of photos and videos. RNNs, on the other hand, capture temporal correlations in data effectively, making them well-suited to time series tasks. The Long Short-Term Memory (LSTM) approach, a famous RNN version, excels at learning order dependencies in sequence prediction problems. Graph Neural Networks (GNNs) is a type of neural network created specifically for graph-structured data, where nodes represent entities and edges represent relationships (Battaglia et al., 2018). While FNNs, CNNs, and RNNs are generally developed for Euclidean data, GNNs are specifically designed for DL with non-Euclidean data structures.

A frequently employed variant of CNN, resembling the structure of Multi-Layer Perception (MLP), comprises multiple convolutional layers followed by sub-sampling (pooling) layers, culminating in Fully Connected (FC) layers. In a CNN model, the input i for each layer is arranged in a three-dimensional format: height, width, and depth, denoted as $n \times n \times k$, where both height n and width are equivalent. The depth corresponds to the channel number, for instance, in an RGB image, the depth k equates to three. Each convolutional layer is equipped with several kernels (filters), denoted as c, which also possess three $s \times s \times b$, akin to the input image. Here, s must be smaller than n, while b can either equal or be smaller than k. These kernels establish local connections and share similar parameters bias β_c and weight ω_c to generate c feature maps h_c of size $n - s - 1$. The convolution layer computes the dot product between its input and weights in Eq. (3), similar to NLP (Natural Language Processing), albeit operating on smaller regions of the initial image. Subsequently, applying a nonlinearity or activation function to the convolution-layer output yields the following result:

$$h^c = f(\omega^c * x + \beta^c) \tag{3}$$

The subsequent step involves down-sampling each feature map within the sub-sampling layers. This action effectively reduces network parameters, thereby expediting the training process and addressing overfitting concerns. For all feature maps, a pooling function (max or average) is applied to adjacent areas of size $d \times d$, where d signifies the kernel size. Finally, the FC layers receive mid- and low-level features, synthesizing high-level abstractions akin to a conventional neural network's last-stage layers. Classification scores are generated through the final layer (e.g., SVMs or softmax). For any given instance, each score represents the probability associated with a specific class (Alzubaidi et al., 2021).

Deep Belief Networks (DBNs), error-feedback Recurrent Convolutional Neural Networks (eRCNNs), Fully Convolutional Neural Networks (FCNs), and Spatio-Temporal Graph Convolutional Neural Networks (STGCNs) are examples of neural networks that have undergone extensive development and have been extended with various architectural variations. DBNs, which can be described as a layered structure of Restricted Boltzmann Machines (RBMs), are one example of such an extension (Yuan et al., 2022). DBNs are made up of visible and hidden units that constitute a two-layer network concept. These several ANN design extensions highlight the variety of deep learning options and approaches. Understanding the different types of neural networks and their specific capacities is crucial for understanding the diverse DL methodologies utilized in traffic management in the context of this inquiry.

The goal of this survey article is to thoroughly analyse the cutting-edge methodology, technologies, and strategies used in traffic management. The study will examine a wide range of subjects, including conventional methods, cutting-edge tactics, intelligent transportation systems, and developing market trends. This survey article seeks to provide a thorough overview of the existing landscape and future directions in traffic management by analysing the benefits, drawbacks, and practical implications of various management strategies.

3. Related work

Manibardo et al. (2021) investigate how well deep learning approaches work for predicting traffic on roads. On traffic datasets, the authors assess how well three deep learning models LSTM, CNN, and a mixture of the two perform in comparison to more conventional machine learning models. They deduced from the experimental findings that shallow modelling techniques frequently outperform deep learning models in terms of performance. Additionally, they outline research opportunities for the given approaches to these problems, some of which are motivated by recent developments in data-based modelling. Among these is the requirement to go beyond accuracy and take into account additional factors that support the actionability of traffic forecasts. The use of generative models for producing realistic traffic data will open up further potential for data augmentation, and they foresee that a centralized repository for traffic data will enable researchers to use the same traffic datasets and replicate discoveries from the literature.

Fan et al. (2020), they provide an overview of recent advances in deep learning for intelligent traffic sensing and prediction, as well as identify future challenges in the field. The authors discuss the use of deep learning techniques such as CNNs, RNNs, and Generative Adversarial Networks (GANs) for various applications in traffic sensing and prediction. They also review datasets commonly used in the field and discuss challenges related to data collection, pre-processing, and modelling. The paper highlights the potential of deep learning for improving traffic management and safety, while also emphasizing the need for further research to address challenges related to data such as multi-source data, large-scale data, fine-grained traffic predictions, real-time data, data privacy, open-source data, and data storage.

In Ali and Mahmood (2018) study, they present a comprehensive review of the current state of research on using deep learning techniques for short-term traffic flow prediction. The authors analyse various studies that have used deep learning models such as CNNs, LSTM, and DBN, and they examine the impact of different factors such as input data, network architecture, and performance metrics on the accuracy of the models. According to the authors, DBN and LSTM are the most popular used. The paper also identifies current challenges related to events in the traffic flow field, such as road accidents, weather changes, and maintenance work. Overall, the authors conclude that deep learning techniques hold great potential for short-term traffic flow prediction and recommend future research to further improve the accuracy and effectiveness of these models such as combining LSTM and CNN and using a Bayesian network model to incorporate the impact of such traffic affecting events with other deep architectures.

In Shi et al. (2019), the research paper provides a review of recent literature on hybrid deep learning techniques for traffic flow prediction. The paper focuses on studies that combine DNN such as Stacked Learners+FCN, CNN+RNN, CNN+RNN+Attention, Embedding+CNN+RNN, and Multi-task Learning. The authors analyse the advantages and limitations of these hybrid models and discuss their performance on various traffic datasets. In addition, they review the various types of data sources used in hybrid deep learning such as historical traffic flows, onboard device OBD data, weather data, social events, speed, journey time, and accident data. Then, they also highlight the importance of feature selection and data pre-processing for the accuracy of the models. The review concludes, that for advance employing both temporal and spatial dependencies in hybrid DNN models for traffic flow prediction becomes more and more complicated. Finally, the authors ordered the hybrid DNN models for traffic forecasting from simple models to complex ones.

In Miglani and Kumar (2019), the authors present a comprehensive review of deep learning models for traffic flow prediction in autonomous vehicles. The authors analyse various studies that have used deep learning models such as CNNs, LSTM networks, and Autoencoders; and discuss the advantages and limitations of these models. In addition, the authors also highlight the importance of real-time processing and scalability for traffic flow prediction in autonomous vehicles. The authors further identify several research challenges, including defining metrics for the evaluation of traffic flow prediction models, unusual factors, data heterogeneity, data imbalance, and limited data availability; and propose possible solutions to address these challenges. Overall, the paper provides a valuable overview of the state of research on deep learning models for traffic flow prediction in autonomous vehicles and suggests potential avenues for future research in the field.

The goal of Yin et al. (2021) study is to present a thorough overview of deep learning-based approaches in traffic prediction from many angles. In particular, they provided a taxonomy and a summary of the traffic forecasting techniques that were already in use in the literature. The most recent techniques used in various traffic prediction applications were then listed. Then, to assist other researchers, they gathered and arranged commonly used public datasets from the literature. They have performed extensive tests to examine the effectiveness of various approaches on a real-world public dataset, providing an evaluation and analysis. Finally, they discussed open challenges in deep learning in the traffic prediction field such as few shot issues, knowledge graph fusion, long-term and real-time prediction, multi-source data, models interpretability, high dimensionality, prediction under perturbation, and choosing an optimal network architecture.

This paper (Mahardhika and Putriani, 2023) reviews the potential use of Artificial Intelligence (AI)-enabled electric vehicles (EVs) in managing traffic congestion. The authors discuss various applications of AI algorithms in optimizing EV routing and charging, managing traffic flow, and collecting data on traffic conditions. They acknowledge the potential benefits of using AI-enabled EVs in reducing the impact of traffic congestion on the environment and improving transportation efficiency. However, the authors also highlight the need for robust data infrastructure and effective AI algorithms to address the challenges in implementing this technology. In the end, the study concludes that AI-enabled EVs have great promise in managing traffic congestion, but continued innovation and investment are necessary for their successful implementation.

In Akhtar and Moridpour (2021), the study provides a review of the use of Artificial Intelligence (AI) in predicting traffic congestion. The authors discuss the different types of AI algorithms used in traffic congestion prediction, including machine learning, neural networks, and fuzzy logic. They also review various data sources that can be used for congestion prediction, such as Global Positioning System (GPS) data, traffic flow data, and social media data. The paper concludes that the use of AI in traffic congestion prediction can provide accurate and timely information to traffic management agencies and help to improve traffic flow. However, the authors highlight the need for further research in this area to improve the accuracy and reliability of congestion prediction models. They pointed out that research should pay attention to real-time traffic congestion estimation problems. Also, they propose another direction for future research to focus on the level of traffic congestion. Finally, an excellent future direction is to give attention to more than one parameter and combine the results during congestion forecasting to make the forecasting more reliable.

A comprehensive survey of the use of Deep Learning (DL) techniques in detecting, predicting, and alleviating traffic congestion, papers covered in this survey were published in the period 2016–2021 (Kumar and Raubal, 2021). The authors review various DL algorithms, such as CNNs, RNNs, and reinforcement learning, and their applications in congestion detection, prediction, and alleviation. They also discuss different data sources used in DL-based traffic analysis, such as traffic flow data, GPS data, and social media data. The paper concludes that DL has great potential in improving traffic management and reducing congestion, particularly in real-time traffic analysis and prediction. However, the authors also highlighted the importance of data standardization, potential synergies with other domains of work, and the potential synergies between the simulation-based and deep learning approaches for traffic prediction tasks.

In Rasheed et al. (2020), the study provides a review of the use of Deep Reinforcement Learning (DRL) in traffic signal control. The authors discuss the advantages of using DRL algorithms in optimizing traffic signal timing, including learning from experience and adapting to changing traffic conditions. They review various DRL algorithms used in traffic signal control (TSC), such as Q-learning, Deep Q-Network (DQN), and Policy Gradient (PG) methods. The paper also highlights different state-of-the-art studies that have applied DRL algorithms in traffic signal control. Then, the paper addresses open issues that can be pursued in this topic in the future such as (i) the effects of dynamicity on DQN, (ii) the learning efficiency of DQN for TSC, (iii) the effects of traffic disturbances on learning in DQN for TSC, (iv) the safety issue of DQN for TSC, (v) the fairness and prioritized access issue, (vi) developing traffic simulators for investigating DQN-based TSCs, and (vii) conducting a literature review of DRL from the TSC perspective.

In Razali et al. (2021), the paper present an assessment of recent research in traffic flow prediction utilizing machine learning and deep learning approaches. The review found a gap in the lack of computationally efficient methodologies and algorithms, as well as, a scarcity of high-quality training data, due to the complicated association characteristics between road sections and congestion patterns in traffic zones or busy locations. The study also discovered a failure to apply dynamically acquired spatial and temporal interactions in deep learning. Despite these obstacles, several methodologies, models, and tactics have been presented to increase traffic flow prediction accuracy. Machine learning approaches such as CNN, LSTM, and others have demonstrated promising results in improving the performance of the proposed methodology. Other literary sources and studies mentioned in more databases, as well as, trials using traffic data recorded in diverse local urban districts, could be added in future studies to provide wider data patterns for model training. Finally, to ensure the progress of the

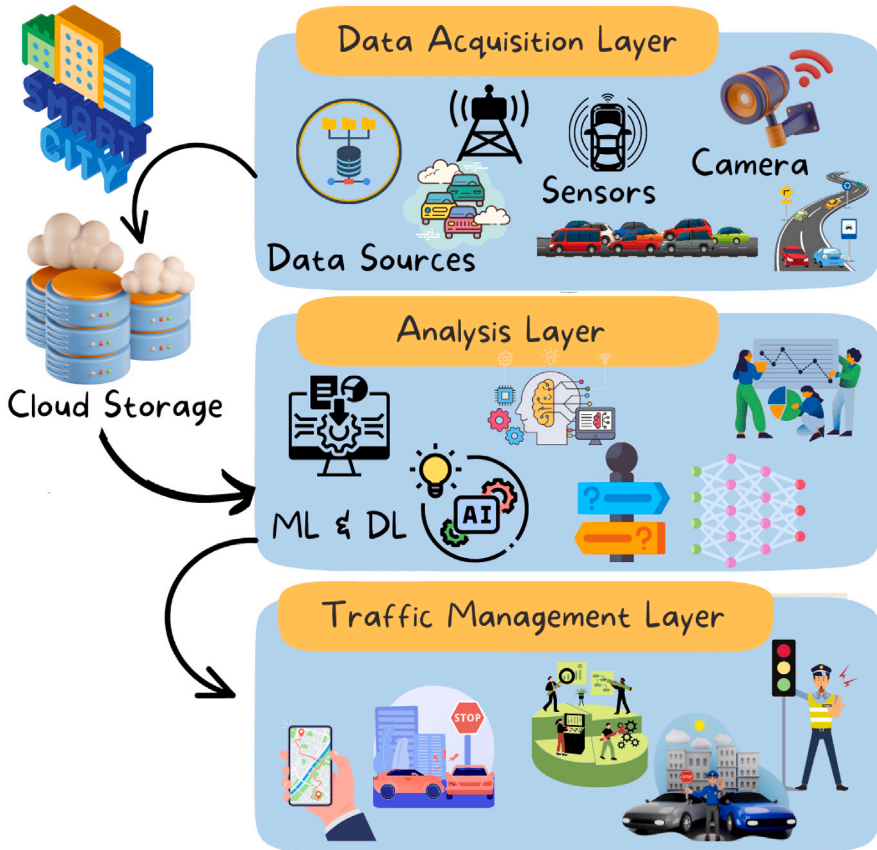


Fig. 1. Traffic congestion managements framework architecture.

Intelligent Transportation System (ITS) technology in smart local urban cities, a framework to solve cybersecurity-related issues for ITS in smart local urban cities must be established.

In Iftikhar et al. (2023), the study covers a comprehensive review of the implementation of deep learning approaches for target detection using Unmanned Aerial Vehicles (UAVs). The study focuses on four critical challenges in target detection, including small-size objects, target occlusion, complex background, and joint issues. The study provides an overview of different deep learning approaches utilized for target detection, including one-stage detectors and two-stage detectors. The study found that one-stage detectors are widely used in target detection due to their fast detection rate, while two-stage detectors are not commonly used due to their low detection rate. Finally, the study identifies some significant challenges and potential research directions for deep learning and target detection. These include (i) improving recognition and detection encryption, (ii) reducing false alarms through the use of multiple visual sensors and stereo vision, (iii) proposing more efficient approaches to reduce power consumption, (iv) optimizing algorithms using laser facts and depth maps, (v) improving the YOLOv4_Drone approach for real-time implementation, (vi) replacing the current design of YOLOv3 with the latest version of YOLOv4 and an online DeepSORT target tracker with a multi-object tracker, (vii) generating high-quality datasets for aerial photography, (viii) merging one-stage and two-stage detectors, (ix) upgrading the detection speed of faster R-CNN, and (x) changing portable datasets of high quality into vast datasets.

Unlike recent research surveys presented in the literature that focus on traffic flow prediction, traffic congestion management, traffic signal control, or target detection using UAVs; our survey focuses on traffic management. In this work, we present a thorough understanding of the benefits, drawbacks, and practical implications of leveraging machine learning and deep learning in traffic management. The study evaluates 20 recent representative traffic management research prototypes that have been published since 2019. Based on the evaluation results, we highlight the current open issues and upcoming directions in the traffic management field while leveraging the potential of machine learning and deep learning techniques.

4. Traffic management framework architecture

To understand the benefits and drawbacks of machine learning and deep learning techniques in traffic management, we propose a generic traffic management architecture and distinguish a set of assessment criteria to evaluate recently proposed research prototypes. The results of this evaluation will help us in identifying possible open issues in the field of traffic management. Fig. 1 depicts a generic architecture for traffic management that we propose in this section.

4.1. Layers of traffic management framework architecture

A typical traffic management system consists of three primary layers including (i) the *Data Acquisition Layer*, (ii) the *Data Analysis Layer*, and (iii) the *Traffic Management Layer*. Each layer in the system has a specific role, as detailed below:

1. Data Acquisition Layer: This layer is in charge of gathering and pre-processing data from numerous sources, such as traffic cameras, GPS data from vehicles, and road sensors. This layer often includes criteria such as data sources, data types, data size, data dimensions, and area type. The data can be stored in the cloud and used later in the data analysis layer. The data acquisition layer is essential for ensuring that the data is appropriately prepared for analysis by the system's subsequent levels.

2. Data Analysis Layer: This layer is in charge of processing and evaluating the information gathered by the data acquisition layer. This layer criteria comprises a variety of approaches, models, parameters, and focus; that are used to derive insights and anticipate traffic flow and congestion. The data analysis layer is crucial for identifying areas of congestion and devising ways to more efficiently regulate traffic flow.

3. Traffic Management Layer: This layer is in charge of developing strategies and making decisions about how to manage traffic flow and congestion based on the insights and forecasts given by the data analysis layer. Decision-making processes, performance evaluation criteria, and technologies utilized to implement traffic management techniques are all part of this layer criteria. The traffic management layer is crucial for putting strategies generated during decision-making processes into action and monitoring their efficacy using performance evaluation criteria.

To summarize, the data acquisition layer is in charge of collecting and pre-processing data, the data analysis layer is in charge of processing and analysing the data, and the traffic management layer is in charge of developing strategies and making decisions about how to manage traffic flow and congestion based on the insights and predictions generated by the data analysis layer. These layers, when combined, comprise a comprehensive traffic management system that can assist in reducing congestion, improving safety, and increasing overall transportation network efficiency.

4.2. Criteria for evaluating traffic management

For each layer of the proposed generic traffic management architecture, we identified a collection of criteria for identifying, measuring, and managing traffic in the transportation network detailed below:

4.2.1. Data acquisition layer

The criteria for the data acquisition layer include:

1. Data source: This criterion refers to the origins or locations from which data is collected for machine learning or deep learning traffic management. Platforms, devices, or systems that give data for analysis, modelling, and decision-making. Data from numerous sources are frequently used in traffic management to provide a comprehensive understanding of traffic conditions. Here are some examples of data sources commonly employed in traffic management (Zhu et al., 2018):

- **Traffic Cameras:** These cameras are strategically placed to record real-time visual data on highways. They provide image or video feeds for many purposes such as traffic flow analysis, vehicle detection, and incident detection.
- **Sensors:** To collect real-time traffic data, many types of sensors are mounted in roadways. Loop detectors embedded in the pavement to measure traffic flow, microwave sensors to detect vehicle presence, and other technologies like radar to obtain additional information about traffic patterns and vehicle behaviour are examples of these sensors.
- **GPS and Navigation Systems:** Vehicle navigation systems and GPS devices give the location, speed, and journey time data. A fleet of vehicles' GPS data can be used to study traffic patterns, congestion levels, and trip time estimation.
- **Mobile Devices:** Smartphones and tablets, for example, can provide useful data for traffic management. Users' anonymized GPS or accelerometer data collected by mobile apps or platforms can provide insights into traffic conditions and travel habits.
- **Traffic Management Systems (TMSs):** Existing TMSs, such as Intelligent Transportation Systems (ITSs), give data via their infrastructure. This can comprise information such as traffic signal data, traffic flow data from loop detectors, incident reports, and road network data.
- **Weather Data Services:** Meteorological data such as temperature, precipitation, wind speed, and visibility are provided through weather data services. Weather data, which may be linked to TMSs, is critical for understanding how weather conditions affect traffic.
- **Public Transport Systems:** Data generated by public transportation networks include bus or train schedules, passenger numbers, and real-time vehicle whereabouts. Incorporating public transportation data aids in the coordination of traffic signal timing and the optimization of transit operations.
- **Social Media and Crowdsourcing:** During traffic incidents or events, social media platforms, and crowdsourcing applications can be significant sources of real-time information. Users can post updates, images, or videos concerning road conditions, accidents, or traffic congestion.
- **Historical Data Archives:** Historical data archives, which are kept by transportation agencies or research institutions, allow access to previously acquired traffic data from diverse sources. These archives are rich in data that can be used to train and validate deep learning models (Miglani and Kumar, 2019; Akhtar and Moridpour, 2021).

2. Data type: Is referred to as the format or representation of the data gathered and used for analysis and model training. The nature of the information being gathered and processed can influence the data type. Here are some examples of data formats commonly used in traffic management (Jiang and Luo, 2022):

- **Numeric Data:** It is often used in traffic management to represent quantitative data. Data such as vehicle speed, traffic volume, occupancy, density, time durations, and distances can be included. Numeric data is frequently utilized in statistical analysis, modelling, and prediction.
- **Image and Video Data:** It represents visual data collected by traffic cameras or other imaging equipment. Vehicle identification, object recognition, and traffic flow analysis all employ these data types. Pixel values or image/video file formats (e.g., PNG, JPEG, MPEG) are commonly used to represent image and video data.
- **Textual Data:** Textual information such as traffic event descriptions, road signage, traffic regulations, and textual properties connected with traffic-related items are included. Natural Language Processing (NLP) approaches can be used to process textual data for tasks such as sentiment analysis, information extraction, and text classification.
- **Geospatial Data:** It represents data related to geographical locations. GPS coordinates, road network topology, boundaries, and other spatial features are examples of data that can be included. Coordinate systems (e.g., latitude and longitude) or spatial data formats (e.g., GeoJSON, Shapefile) are commonly used to represent geospatial data.
- **Time Series Data:** It is a collection of data points gathered over time. Time series data in traffic management might comprise traffic flow measurements, vehicle speed data, and other time-dependent properties. Time series data is defined by timestamps or time intervals and is used to analyse trends, forecast, and find anomalies (Xing et al., 2022; Xiao et al., 2023).

3. Data size: This refers to the amount of data acquired from each source, which can vary greatly based on the application, project scope, and data sources employed. Because of the large volume of data acquired from diverse sources, data size in traffic management might be enormous.

4. Data dimensions: It refers to the various qualities or characteristics of data collected and used for traffic control. Here are some examples of frequent data dimensions in traffic management data acquisition (Jiang and Luo, 2022):

- **Spatial Dimension:** Traffic data frequently includes a spatial component that represents the physical location or position of traffic-related events. GPS coordinates, road network topology, traffic flow information, and geographical borders are instances of this.

- **Temporal Dimension:** Traffic conditions fluctuate over time, and understanding traffic patterns and creating accurate predictions requires capturing the temporal dimension. Time stamps, event timestamps, time intervals, and historical traffic patterns are all examples of temporal data.
- **Traffic Attributes:** Various traffic variables are collected to characterize the state and behaviour of cars and road infrastructure. Vehicle speed, acceleration, occupancy, density, traffic volume, vehicle type, and lane statistics are all included.
- **Environmental Factors:** Environmental factors can have a considerable impact on traffic patterns. Capturing information such as weather conditions (temperature, precipitation, visibility), road conditions (e.g., wet, dry, ice), and other pertinent aspects that can influence traffic flow is one example of data acquisition.
- **Sensor Data:** To gather real-time traffic information, data acquisition systems frequently employ several types of sensors. Data from traffic cameras, loop detectors, microwave sensors, radar, and other sensor technologies may be included.
- **Event Data:** Accidents, road closures, construction zones, and special events can all have a substantial impact on traffic flow. Data collection may entail gathering information about these events, such as their location, length, and severity.

5. Type of area: This is the geographical area or location where the traffic data is collected and processed. Traffic patterns, characteristics, and management tactics can all be influenced by the type of region. Here are some instances of common traffic management areas:

- **Urban Areas:** High population densities, complicated road networks, and substantial traffic volumes characterize urban areas such as cities or metropolitan regions. In metropolitan settings, traffic management is frequently focused on congestion reduction, traffic signal optimization, and pedestrian safety.
- **Rural Areas:** In comparison to urban regions, rural areas have lower population densities, less complex road networks, and often lower traffic volumes. Rural traffic management may concentrate on road safety, boosting connectivity among rural settlements, and solving special issues such as animal crossings or agricultural vehicle movements.
- **Highways and Expressways:** Highways and motorways are important road arteries built to accommodate long-distance travel and high-speed traffic. Highway traffic management entails monitoring traffic flow, detecting issues or accidents, and optimizing lane utilization and toll-collecting efficiency.
- **Residential Areas:** Residential regions are predominantly residential areas with reduced traffic volumes and speed limits. Residential traffic management emphasizes safety measures such as traffic calming, speed limit enforcement, and assuring pedestrian and cycling safety.
- **Special Event Zones:** Special event zones are areas where temporary events or activities take place, such as sports events, concerts, or festivals. Traffic management in special event zones requires planning for increased traffic demand, managing road closures, and ensuring smooth ingress and egress for event attendees.

In summary, the data acquisition layer collects and pre-processes data from numerous sources, such as cameras, sensors, and GPS devices. It must take into account the data's kind, size, dimensions, and geographic location to ensure that it is correctly prepared for analysis by higher layers of the system.

4.2.2. Data analysis layer

This layer is in charge of processing and analysing the data acquired and pre-processed from the data acquisition layer. This layer covers multiple criteria such as techniques, models, parameters, and focus that are utilized to extract insights and anticipate traffic flow and congestion (Razali et al., 2021; Ranjan et al., 2023).

1. Techniques: There are many ways available for data analysis in TMSs. Statistical analysis, machine learning, and deep learning are examples of these. Based on historical data, machine learning, and deep learning can be used to create predictive models that forecast future traffic conditions.

2. Models: Are mathematical representations of data that can be used to generate predictions or extract insights. Regression models, time series models, neural networks, and other machine learning or deep learning models may be used in traffic management. These models can be trained on historical data to forecast future traffic conditions, such as the likelihood of congestion on a specific route stretch. Decision trees, support vector machines, and random forests are examples of machine learning approaches that could be utilized in this layer. CNN, LSTM, and RNN are instances of deep learning approaches.

3. Parameters: The parameters utilized in the data analysis layer are derived from datasets acquired using various practices such as cameras, sensors, or other data collection mechanisms. These datasets vary by country; however, they frequently include traffic factors such as traffic volume or vehicle count, traffic speed, time, and average number of vehicles or traffic density (Razali et al., 2021).

4. Focus: The data analysis layer's focus is determined by the system's specific aims. The goal of a traffic analysis system may be to predict traffic patterns, detect congestion hotspots, or optimize traffic flow.

In essence, the data analysis layer is in charge of processing and evaluating the information gathered from the data acquisition layer. As a result, the system may identify regions of congestion and devise methods for more effective traffic management.

4.2.3. Traffic management layer

This layer is responsible for using the insights and predictions generated by the data analysis layer to develop strategies and make decisions about how to manage traffic flow and congestion. This layer includes decision-making, performance evaluation, and technologies used criteria (Oladimeji et al., 2023; Ata et al., 2019).

1. Decision-making: Consists of the parameters that are considered while making traffic-management choices. These factors may include traffic volume, speed, trip time, and congestion levels. To control congestion, the system may be designed to devise tactics such as rerouting traffic, altering traffic lights, or introducing road closures.

2. Performance evaluation: This criterion includes the measures that are used to assess the success of traffic management strategies. Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) are instances of used measurements. Other measures including accuracy, precision, recall, and F1-score, are used to contrast various traffic flow forecasting methods. The traffic management layer can determine the accuracy of their system by evaluating system performance.

3. Technologies: This criterion refers to technologies employed, specific tools, and techniques used in the implementation of traffic control plans. ITS, traffic signal control systems and the IoT are examples of such technologies. The system may alter its methods based on data from sensors and cameras. Furthermore, object detection technologies such as YOLO are used, which increases the possibilities in this sector.

In summary, the traffic management layer is in charge of developing strategies and making decisions about how to manage traffic flow and congestion based on the insights and forecasts given by the data analysis layer. The system applies measures to manage traffic more effectively after identifying regions of congestion.

5. Research prototypes

This section provides a comprehensive analysis of 20 representative traffic management research prototypes. These research prototypes include a variety of studies that have undergone quantitative analysis for a full examination. The research prototypes were chosen to represent a

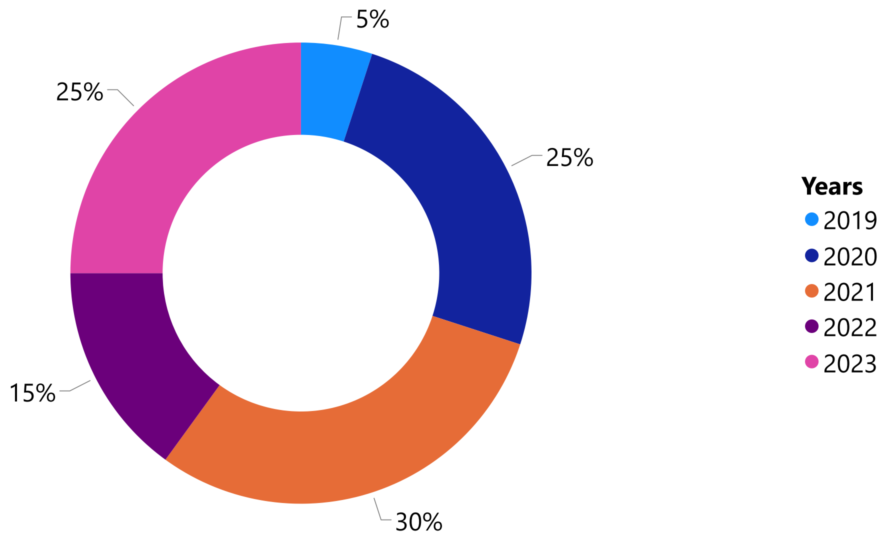


Fig. 2. Distribution of articles in publishing years.

wide set of Machine Learning (ML) and Deep Learning (DL) techniques used in the subject of traffic management. We ensure a rigorous and objective assessment of their findings and consequences by exposing them to quantitative analysis. This analysis provides a more in-depth understanding of traffic management advancements, obstacles, and prospective solutions.

5.1. Research strategy

A systematic search technique was used to identify and choose relevant research, to capture the junction of traffic management, ML, and DL methodologies. To ensure a thorough examination of relevant material, the following strategy was used:

1. Databases Used: A thorough search was undertaken on databases such as PubMed, IEEE Xplore, Scopus, and Web of Science, which were chosen for their coverage of multidisciplinary research in traffic management, machine learning, and deep learning.

2. Keywords and Search Terms: A deliberate keyword selection was used, which included terms like "traffic management", "machine learning", "deep learning", "traffic prediction", "traffic flow optimization", "public transportation management", "emergency management", and "smart cities".

3. Search Filters and Criteria: The search was restricted to studies published after 2019 to emphasize recent breakthroughs in the field. Language (English), publication type (peer-reviewed articles and conference papers), and applicability to traffic management using ML or DL approaches were also considered.

4. Search Strings Example: Boolean operators were used to produce search strings such as ("Traffic management" OR "Traffic prediction" OR "Traffic flow optimization" OR "Public transport management" OR "Emergency management" OR "Smart cities") AND ("Machine learning" OR "Deep learning").

A diverse tendency can be seen in the distribution of articles covered in this study over publishing years. For example, from 2019 to 2021, publications increased gradually, peaking at 30% in 2021. From 2022 to 2023, publications decreased somewhat to 15% and then increased once more to 25% in 2023 as shown in Fig. 2. Of the selected articles, 95% of them were published in journals, which makes up the bulk of the publication venues; only 5% was presented at a conference proceeding. Among the journals and conference proceedings, MDPI represented the largest share with 25%, followed by IEEE with 20%. ScienceDirect contributed 15% of the publications, whereas arXiv preprint, Hindawi, and PeerJ Inc. each hosted 10%, 5%, and 5% of the articles, respectively. Additionally, Indian Society for Education and Environment, The Science and Information Organization, Ahmedabad: Solaris Publication, and Tamilnadu: Fast Track Publications individually represented 5% of the selected articles as shown in Fig. 3. This analysis, with variable degrees of contribution across different journal platforms, indicates a significant preference for journals over conferences in the publication of these publications.

5.2. Overview of major traffic management research prototypes

To provide a comprehensive understanding of the major research efforts in the field of traffic management, we present an overview of several impactful studies. These studies have contributed significantly to the advancement of traffic management strategies by addressing various aspects and challenges. By presenting this overview, we aim to shed light on the breadth and depth of research conducted in the field, encompassing a wide range of topics, machine learning or deep learning techniques, and findings. This exploration allows us to gain valuable insights into the latest trends, innovations, and key findings that researchers have uncovered in their pursuit of more efficient and effective traffic management solutions. Through this analysis, we can identify important themes, potential gaps, and avenues for future research, fostering continued progress in the field of traffic management.

In Balasubramanian et al. (2023), an IoT system that uses machine learning to manage traffic and detect accidents in smart cities was proposed. The system includes sensors, edge devices, and cloud-based servers to collect and analyse data. The study shows that the proposed system effectively detects accidents and manages traffic, demonstrating the potential of machine learning for improving transportation systems in smart cities. Potential benefits noted in this study include: (i) innovative approach design: to improve traffic management and accident detection in smart cities, the study presents a ground-breaking approach that combines machine learning, the Internet of Things, and secure data transfer, and (ii) notable effects: it is projected that the proposed system will improve overall travel safety and experience while concurrently decreasing travel time, traffic congestion, accident frequency, and environmental pollution. In addition, the study identifies limitations of the proposed system, such as the need for a reliable internet connection, high implementation and maintenance costs, and the potential for false alarms. The system's effectiveness may also vary depending on the specific context of the smart city where it is deployed.

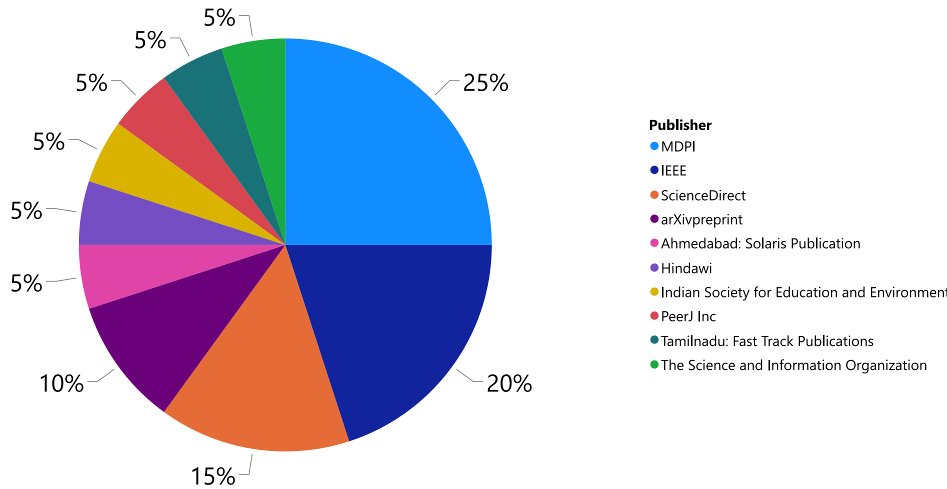


Fig. 3. Distribution of articles by publisher.

The study (Lilhore et al., 2022), discusses the challenges of traffic congestion and the potential benefits of using IoT and machine learning for Intelligent Traffic Management (ITM). The proposed Adaptive Traffic Management (ATM) system uses sensors embedded in vehicles and infrastructure to collect and analyse traffic data in real-time. The system utilizes machine learning algorithms to optimize traffic signal schedules and reduce congestion. The results of the study show that the proposed ATM system outperformed conventional traffic-management strategies and has the potential to improve transportation planning in smart cities. The advantages of this work include: (i) the use of ML and IoT technology, the suggested ATM system can lessen traffic congestion, increase road safety, and improve transportation efficiency and (ii) based on the projected movements and traffic volume, the proposed ATM system may adjust traffic light timings to various traffic circumstances. While effective in reducing congestion and improving traffic flow, limitations of the system include the need for a stable internet connection and high costs for implementation and maintenance. Overall, the study appears to be well-conducted, and its findings could have practical applications in the development of intelligent transportation systems.

In Kothai et al. (2021), the authors state that the proposed hybrid deep learning model, which combines Convolutional Neural Networks (CNN) and Boosted Long Short-Term Memory Ensemble (BLSTME), effectively predicts traffic congestion in smart cities and can reduce delays, energy consumption, and travel management for passengers. The model uses techniques such as density computation, edge calculation, dilation, thresholding, frame differencing, and contour detection to predict traffic flow and detect the vehicle zone. The model is compared with existing models and achieves more than 10% higher accuracy, recall, and precision in predicting traffic collisions. The authors suggest that future research could focus on the hybrid incorporation of predictors to improve collision prediction in the network. The paper states that the following are the strengths of the hybrid deep learning algorithm that is suggested for wide traffic congestion prediction in smart cities: (i) it can handle the dynamic behaviour of the vehicles in the network and get around the overfitting issue, (ii) it can effectively train the classifiers using BLSTME after extracting both spatial and temporal information from the traffic photographs, and (iii) it can outperform current models like the autoencoder, ConvLSTM, and PredNet in terms of accuracy, precision, and recall. However, it is important to note that the study has limitations, such as the use of a simulated traffic scenario instead of real-world data and the absence of a comparison with other state-of-the-art deep learning models. Therefore, further research is needed to validate the effectiveness of the proposed algorithm in real-world traffic scenarios and to compare it with other existing models.

The proposed Video-based Intelligent Transportation System (V-ITS) in Abbas (2019) appears to be an efficient solution for predicting illegal activities during highway driving. The study utilizes deep learning algorithms including CNN and Deep Belief Network (DBN) to classify vehicle illegal activities in multiple video frames and achieve robust results without delay. The evaluation results indicate that the V-ITS system outperforms state-of-the-art video processing systems. The work of the paper has several advantages, including (i) the system can lessen the negative effects of transportation on the environment by incentivizing drivers to stick to the speed limit and refrain from making pointless lane changes, (ii) the system can extract and categorize characteristics from video frames by utilizing cutting edge deep learning techniques and the already existing video surveillance infrastructure, and (iii) the system can achieve higher detection rates and accuracy than the most advanced video-based vehicle tracking techniques. Overall, the study contributes to the field of intelligent transportation systems, as it addresses the critical issue of tracking illegal activities to enhance traffic safety. However, it would be important to evaluate the system's performance on a wider range of datasets and in different environmental conditions to determine its generalizability and practicality for real-world applications.

The study (Ravi et al., 2021), proposes a machine learning-based traffic management system to analyse traffic flow and predict congestion on specific paths. The system uses video-based vehicle counting, which involves object detection, tracking, and counting using You Only Look Once (YOLO). The proposed system provides a simple, accurate, and early prediction of traffic congestion and can give highly accurate results. This system is suitable for real-world applications, and it can notify vehicles intending to travel on congested paths well in advance. By using this system, free-flowing road traffic can be achieved, leading to faster connectivity and transportation systems in smart cities. The advantages of this study include: (i) utilizing machine learning to evaluate traffic flow and recommend the best routes, the system can lessen traffic congestion and its detrimental impacts, including accidents, pollution, and time wasted and (ii) the technology can identify cars travelling the wrong way and take pictures of them for additional examination and enforcement, improving traffic safety and order. However, the study has some disadvantages, such as the limited scope of the study to a specific location and the need for further testing and validation of the proposed system in different scenarios and environments.

The proposed system architecture in Abhijit et al. (2021) uses video-capturing devices instead of sensors which reduces setup costs. The system's neural network-based object detection algorithm can identify and count different types of vehicles, which helps in calculating the density of vehicles in each lane. The authors propose a vehicle counting model and a dynamic traffic signal timer setting algorithm based on deep learning technology using different versions of YOLO for object detection. The timing algorithm then uses this density data to adjust the green signal timer for each lane, which can help in reducing traffic congestion. Overall, the proposed system architecture appears to be a good solution for optimizing traffic flow in a smart city. The suggested deep learning-based smart traffic control system has the following benefits: (i) dynamically modifying the signal timers in response to real-time traffic data recorded by cameras, it can lessen traffic congestion and enhance traffic flow and (ii) it can reduce waiting times and stoppages at junctions, saving important resources like time and petrol. However, it is important to note that the limitations of the system, such as the processing time and accuracy of the object detection algorithm, should also be carefully considered.

The study in Saleem et al. (2022), proposes a fusion-based intelligent traffic congestion control system for vehicular networks using machine learning techniques. The system aims to predict and control traffic congestion by assembling data from the Internet of Vehicles (IoVs)-enabled Vehicular Networks (VNs). The study reports that the proposed system exhibits 95% accuracy and a 5% miss rate, which are better than previous approaches. The work described in this paper has the following strengths: (i) to reduce traffic congestion in smart cities, the suggested FITCCS-VN uses machine learning (ML) techniques to gather traffic data and route traffic on available routes and (ii) to prevent traffic jams, the proposed system offers drivers cutting-edge services that allow them to see traffic flow and the number of vehicles available on the road remotely. However, the study has some limitations, such as the use of simulation data instead of real-world data, and the need for further improvement in accuracy by using federated learning and Alexnet. Additionally, the study does not provide a detailed analysis of the system's performance in different traffic scenarios, and it does not discuss the feasibility of implementing the system in real-world conditions.

The study in Majumdar et al. (2021), shows that LSTM-based models connected to IoT speed sensors can predict congestion propagation across complex road networks in real-world conditions. The models show predictive accuracy ranging from 84% to 95%, which is good for visualizing congestion propagation on real road networks over short timescales. The results suggest that the use of LSTM methods connected to IoT speed sensors provides a method for traffic congestion prediction in smart cities. The application of LSTM networks to forecast the spread of traffic on road networks using data from IoT sensors are covered in this study. One of the advantages is the use of LSTM networks which are capable of accurately predicting the propagation of congestion and capturing the intricate and dynamic behaviour of traffic flow over time. On the other hand, the limitation of this study is that it only considers short-term congestion prediction over 5-minute intervals. Predicting congestion over longer time horizons may require more sophisticated modelling approaches and additional data sources. Additionally, the study focuses on congestion prediction based on average speed, which may not capture all aspects of traffic flow dynamics. Incorporating additional data sources, such as vehicle trajectories or weather data, may lead to more accurate congestion predictions. Finally, the study only considers a single city and road network. Evaluating the generalizability of the approach across different cities and regions would be an important avenue for future research.

The proposed method of using improved Mayfly Optimization with Light Gradient Boosting Machine (MOLGBM) for smart city traffic prediction appears to be a promising approach (Jenifer and Priyadarsini, 2022). The use of feature engineering, feature selection techniques, and the incorporation of a nature-inspired optimization algorithm, may improve the accuracy and efficiency of the prediction model. The proposed model achieved a 2% improvement in accuracy and showed better performance on evaluation metrics in terms of Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) compared to existing literature. One of the advantages is the suggested IMFO-LGBM algorithm outperforms baseline techniques like LGBM, random forest, and XGBoost in terms of accuracy and error rates. However, it should be noted that the study was only evaluated on a specific dataset and further validation on other datasets and in real-world scenarios is necessary to assess its generalizability and effectiveness.

The research work in Yang et al. (2021), proposes a method for improving the accuracy and speed of vehicle congestion detection in urban intersections using the YOLOv3 algorithm and Lucas-Kanade (LK) optical flow method. The method takes the vertex of the detection box as the image feature point and calculates the global speed, which significantly reduces the calculations compared to the traditional optical flow method. The experiments and analysis show that the proposed method has strong anti-interference ability and good vehicle congestion detection accuracy. The following are a few benefits of their work: (i) high-dimensional and complicated data, like GPS, traffic flow, photos, and videos, can be handled by deep learning and (ii) deep learning is capable of identifying patterns and dynamics in the spatiotemporal domain of traffic congestion by utilizing vast amounts of historical and real-time data.

In the paper (Zheng and Huang, 2020), the authors go over how time series analysis using deep learning may be utilized to improve traffic flow prediction. Moreover, the authors discuss the significance of precise traffic flow forecasting for efficient traffic management. The Autoregressive Integrated Moving Average (ARIMA) model and the Backpropagation Neural Network (BPNN) model, two established forecast models, were used to compare the performance of the authors' Long-Short Term Memory (LSTM) network-based traffic flow forecast model. With a mean RMSE of 14.4438, long-term traffic flow forecast trials utilizing actual traffic flow data demonstrated that the proposed LSTM network surpassed the traditional models in terms of prediction accuracy. The advantages of the work include: (i) based on three measures (MAE, MAPE, and RMSE), the LSTM network beat the ARIMA and BPNN models in terms of prediction accuracy and (ii) the LSTM network can help with traffic management decision-making by disclosing the dynamic evolution law of traffic flow. According to the authors, in their future work, they will concentrate on enhancing the capacity to forecast traffic flow under a variety of atypical conditions that can affect traffic flow, including inclement weather, traffic accidents, and significant events.

In Singh et al. (2023), the study discusses the use of a knowledge graph to model and simulate urban traffic using real-time geo-spatial data. The authors utilized the OpenStreetMap (OSM) and OSMNx packages to source traffic data and applied filtering, feature engineering, and optimization techniques to prepare the data. The model was based on a sequential Recurrent Neural Network (RNN) applied to a Neo4j graph of the road infrastructure. The study identified that the knowledge graph could simulate traffic congestion on a variety of urban road networks using mathematical modelling tools. The contributions of this paper can be summarized as; (i) the development of a knowledge graph-based system that can manage urban traffic, (ii) the demonstration of the capabilities of knowledge graphs in minimizing and optimizing traffic congestion, and (iii) the authentication and verification of simulation experiments using deep learning Artificial Intelligence (AI) models trained on feature-engineered real-time traffic data. Some advantages of the work presented in this paper include: (i) real-time data integration: to extract Riyadh city's road geometry and topology and to update each road segment's traffic condition and congestion level, the article leverages real-time geospatial data from the HERE platform and (ii) tailored and scalable solution: the study shows how the knowledge graph-based traffic simulation may be expanded to encompass vast swaths of the city with millions of nodes and edges, as well as tailored to various scenarios like accidents, events, or detours. However, the study had many limitations such as; (i) the focus on a single city and a specific time period, which may not be representative of traffic patterns in other cities or during other times of the year, (ii) depends on data from OpenStreetMap and other publicly available sources, which may be incomplete or out of date, potentially resulting in mistakes in the traffic model, (iii) used a variety of algorithms and models, such as the LSTM network and the ARIMA model, that may not be optimal for all traffic forecasting scenarios, (iv) only takes into account regular traffic circumstances and does not account for unusual events such as accidents or road closures, which can have a substantial impact on traffic flow, (v) the study fails to take into account the social and economic consequences of traffic congestion (e.g., higher air pollution, longer commuting times, and lower productivity), and (vi) the study does not assess the feasibility of implementing the suggested system, as well as the associated costs and resources required.

In Anjaneyulu and Kubendiran (2022), the authors concentrate on the issue of urban traffic congestion and the role of Sustainable Transport Systems (STSs) in traffic prediction and management. The research offers a hybrid Xception Support Vector Machine (XPSVM) classifier model for predicting Short-Term Traffic Congestion (STTC) every 5 min over an hour. Using the weight regularization technique and a fine-tuned binary hyperplane mechanism, the Xception and SVM classifiers are merged to accurately forecast the output. The dataset utilized in this work was obtained from Google Maps and consisted of pictures captured using the Selenium automation tool from Bangalore, Karnataka, India. The study indicates that accurate short-term traffic congestion forecasting is critical for urban transportation infrastructure design and real-time traffic optimization. The work of the authors has the following advantages: (i) the suggested XPSVM model has high accuracy (97.16%) and low error rate (0.95% and 0.5% for Type 1 and Type 2 errors, respectively) for predicting short-term traffic congestion and (ii) the suggested model employs a hybrid strategy that combines the SVM classifier's classification performance with the feature extraction capacity of the Xception classifier. On the other hand, the study's shortcomings include the small dataset utilized, the need for additional testing with other optimization strategies, and the possibility of replacing the SVM optimizer with a logistic regression classifier to increase the model's robustness and speed.

The research work in Qi and Cheng (2023), presents a deep spatial and temporal network model, namely, Defect detection of Shield tunnel lining based on Graph Convolutional Networks (DSGCN) for anticipating traffic congestion in Chinese cities. The suggested architecture divides the traffic network using grids, with each grid representing an individual region. These grid regions' centroids are abstracted as nodes, and the dynamic correlations between the nodes are expressed as an adjacency matrix. The DSGCN model captures the spatial correlation between regions using a Graph Convolutional Neural Network (GCNN). The DSGCN model also captures temporal correlation using a two-layer long and short-term feature model, namely, Defect detection of Shield Tunnel lining based on Machine learning (DSTM). Experiments on real PeMS datasets show that the proposed DSGCN model outperforms competing baseline models and delivers higher accuracy in traffic congestion prediction. The DSGCN model presented in this research has the following advantages: (i) by utilizing a deep spatial and temporal memory model and a graph convolutional neural network, it can capture both the spatial and temporal characteristics of traffic congestion data and (ii) experiments on real PeMS datasets show that it can surpass the current baseline models in terms of prediction accuracy and error metrics. It is crucial to emphasize, however, that the study had some limitations. For instance, the proposed model may not be relevant to other types of traffic networks or geographical regions. Furthermore, the study merely predicted traffic congestion and did not evaluate viable methods for congestion mitigation. More work is needed to solve these constraints and to investigate the DSGCN model's possible uses in real-world traffic management scenarios.

Researchers in Mohanty et al. (2020), propose a deep learning model based on LSTM neural network architecture to predict traffic congestion in neighbourhoods within a region. The model uses inputs such as congestion scores from earlier times and real-time measures of regional traffic. The suitability of the congestion score for characterizing neighbourhood traffic conditions and its predictability using the inputs are demonstrated using Newell's simplified theory of kinematic waves. The deep learning approach is compared to three benchmark models and shown to be superior. The model is made robust even in adverse settings and demonstrated to be scalable using a pared-down version of the freeway network in the San Francisco Bay Area. The model is further improved by incorporating the route choice of individuals through weighted undirected graphs and learning features through graph convolutions. The following are a few advantages of their work: (i) the model can produce reliable and accurate estimates of neighbourhood congestion scores while capturing the intricate and dynamic patterns of regional traffic, (ii) high-dimensional and correlated inputs can be handled by the model without the need for feature extraction or selection to be done by hand, and (iii) by adding graphical representations of the inputs and using graph convolutions to learn features, the model can be enhanced. However, the limitations of the proposed approach include a lack of data on human behaviour, reliance on a simplified traffic model, limited geographical coverage, lack of real-world validation, and computational complexity. The approach is based on traffic data and does not explicitly consider driver behaviour or large-scale traffic patterns. In addition, the research work does not provide detailed information about real-world validation or data sources and collection methods. The use of deep learning models could also be computationally intensive and limit the scalability of the approach for large-scale applications.

The study in Cheng et al. (2021), proposes a short-term traffic flow prediction framework that integrates econometrics and deep learning. The framework is applied to Yanan urban expressways, and a Vector Auto-Regression (VAR) model is introduced to investigate the predictable relationship of traffic variables. Speed is selected as the response variable, and Traffic Index (TI) and traffic volume are the explanatory variables for short-term traffic flow prediction. A hybrid CNN-LSTM network model is then proposed to forecast short-term speed using historical data on speed, TI, and traffic volume. The study shows that using multi-feature prediction is better than using one feature, and the CNN-LSTM network model using multi-feature prediction is a promising method for predicting the short-term traffic flow of Yanan urban expressways. The study provides insights into elaborative traffic management and visualization guidance for travel information releasing and traffic congestion management. The following are some advantages of the work: (i) the response variable and predictors for the prediction job may be found using the vector autoregression model, which can assess the predicted link among traffic variables, such as speed, traffic index, and traffic volume, and (ii) the hybrid CNN-LSTM neural network model can overcome the shortcomings of the single LSTM and CNN models, such as the long-term memory issue and the spatial feature extraction issue, and fully use the spatiotemporal aspects of the traffic flow data. In contrast, the study's limitation is that the TI used in this study is specific to Shanghai, and its portability to other Chinese cities needs further exploration.

In Utomo et al. (2020), the research work proposes a traffic congestion detection method using a fixed-wing Unmanned Aerial Vehicle (UAV) based on the YOLO deep learning algorithm. The system can detect and classify four types of vehicles, namely cars, motorcycles, buses, and trucks. The method was tested in two experimental scenarios, public video recording and real-time UAV video streaming, and achieved a precision of 90.75% and 90%, respectively. The following are some strengths of this work: (i) real-time video data transmission from fixed-wing UAVs is sent to the control room, where CNN's YOLO architecture is used to identify and categorize vehicles, and (ii) fixed-wing UAVs can help prevent or lessen congestion by determining the level and status of traffic density depending on the portion of the route that is covered by vehicles. However, the study has some limitations, such as the dataset used only contains vertical vehicle images, which may affect the accuracy of detection in certain situations. In addition, the

Table 1
Evaluation of traffic management research prototypes considering the data acquisition layer.

Ref	Data Acquisition Layer				
	Data Source	Data Type	Data Size	Data Dimensions	Type of Area
Balasubramanian et al. (2023)	C, S	I, VL	RTD, SD	-	ATM, SC
Lilhore et al. (2022)	C, S	I, VL	RTD, 100 vehicles, 5000 m road	-	ATM, F
Kothai et al. (2021)	C, Seoul city	I	125 images	S, T	The arterial network of the city
Abbas (2019)	C, GRAM-RTM dataset	VF	RTD, the dataset of 600 traffic violations	S	H
Ravi et al. (2021)	VS	VF	RTD, the dataset of 1024 nodes	S	H
Abhijit et al. (2021)	C, MS-COCO dataset	VF	RTD, the dataset contains around 330,000 images of 80 different classes	S	One-way road
Saleem et al. (2022)	S, Road Traffic Dataset (Kaggle)	Text file (MS Excel)	SD, Instances: 2282 Features: 515 Records: 1048576	-	SC
Majumdar et al. (2021)	S, Traffic data for Buxton, UK.	Text file (MS Excel)	SD, the dataset contains 1015 samples per week	T	Town road
Jenifer and Priyadarsini (2022)	S, Time Series in IoT dataset (Kaggle)	Text file (MS Excel)	SD, the dataset contains 59,930 records	T	Road junctions
Yang et al. (2021)	C, (University lab)	VF	SD, the dataset contains a duration of 10 min	T	Urban intersection road
Zheng and Huang (2020)	S, OpenITS	TS	SD, the dataset contains 960 samples with 15min intervals	T	Urban Road
Singh et al. (2023)	S, (OpenStreetMap) OSMNx	TD (text)	RTD, the dataset contains 24,720 nodes and 29,097 relationships	S, T	Urban road networks
Anjaneyulu and Kubendiran (2022)	C, Google Maps with the Selenium tool	I	SD, the dataset contains 1 GB of data, and 2500 images	T	Urban road
Qi and Cheng (2023)	S, PeMS	TD (Graph)	SD, the dataset contains 141 nodes, 5 min of traffic data	S, T	H, F
Mohanty et al. (2020)	S, Freeway/ Highway in the San Francisco Bay area	TD	SD, the dataset contains 54 neighborhoods	T	H, F
Cheng et al. (2021)	S, Traffic police corps of shanghai public security bureau	TS	SD, 7 roads segments, Total length 11.16 km	S, T	Urban expressways
Utomo et al. (2020)	C, YouTube video, fixed-wing UAV	VF	RTD, the dataset has 6000 of vehicle objects	S	H
Abdullah et al. (2023)	S, and mobile phones (PEMs dataset of the United States)	TD	SD, the dataset contains 200 countries, GPS, mobile phone, traffic sensors	T	H
Li et al. (2020)	S, relational database, NoSQL data storage, unstructured data storage	TS, TD	SD, the dataset contains, records that exceeds 300TB	T	H
Mihaita et al. (2020)	S, RTM, Motorway in Sydney Australia	TS (Time-interval)	SD, the dataset contains, 36.34 million data points	S, T	a long motorway

Data Source	Data Type	Data Size	Data Dimensions	Type of Area
C Camera S Sensors RTM Road-Traffic Monitoring VS Video Sequence	I Images VL Vehicle Locations VF Video Frames TS Time-series TD Traffic Data	RTD Real-Time Data SD Static Data	S Spatial T Temporal	ATM Active Traffic Management SC Smart City F Freeway H Highway

study was conducted only on roads outside cities, and its applicability in urban environments needs to be further investigated. Furthermore, the study did not consider the impact of weather conditions, such as heavy rain or fog, which may affect the accuracy of the detection.

The paper in Abdullah et al. (2023), proposes deep learning to forecast traffic indicators, such as speed, weather, flow, and accident likelihood, to improve traffic management in smart cities. The proposed approach utilizes historical and real-time information from various sources, such as social media, weather information, and road conditions information, stored in edge servers. The traffic data is extracted using Bidirectional Recurrent Neural Networks (BRNNs) with soft Gated Recurrent Units (GRU) and classified into congested and non-congested classes. The proposed approach has the potential to enhance the precision and effectiveness of traffic forecasting, which can be helpful in various applications such as route planning, traffic management, and congestion mitigation. The following are some advantages of this work: (i) to increase the precision and effectiveness of traffic forecasting, the method can extract many variables from a variety of information sources, including social media, weather, traffic, and road conditions, and (ii) the method can use an optimization technique to optimize the DL architecture's hyperparameters based on historical and real-time traffic data. On the other hand, the study does not present any empirical evidence or experimental results to validate the effectiveness of the proposed approach.

The research in Li et al. (2020), proposes a framework for traffic congestion called Machine learning and Fuzzy comprehensive evaluation-based framework for Traffic Congestion Prediction and Visualization (MF-TCPV), which includes a raw data layer, data processing layer, and data presentation layer. The proposed deep learning model called Long Short-Term Memory (LSTM) - Spark Parallelization Relevance Vector Machine (SPRVM) is used to predict traffic congestion and is shown to be more accurate than other methods, achieving a prediction accuracy of 92.36%. The research suggests further improvements such as considering more factors affecting traffic flow, optimizing the parameters with complex heuristic algorithms, and exploring the causes of congestion by constructing a traffic congestion cause–effect diagram. The framework has been verified using real data from Whitemud Drive in Canada. The work described in this paper has the following potential advantages: (i) the MF-TCPV framework can forecast and visualize traffic congestion in the future based on a variety of parameters, including average speed, road occupancy, and traffic flow density, and (ii) the framework implements the visualization of congestion prediction findings and the integration of multi-source heterogeneous traffic data using DataX and DataV. In contrast, the study has several limitations

Table 2
Evaluation of traffic management research prototypes considering the analysis layer.

Ref	Analysis Layer			
	Technique	Model	Parameters	Focus
Balasubramanian et al. (2023)	ML	DBSCAN	VD, RTL, AL	CTC
Lilhore et al. (2022)	ML	DBSCAN	TV	ED
Kothai et al. (2021)	DL	CNN, BLSTME, AdaBoost	TD	ED
Abbas (2019)	DL	CNN, DBN	VS	VIA
Ravi et al. (2021)	ML, DL	YOLO CNN	TF, TD	CTC
Abhijit et al. (2021)	DL	CNN, YOLO	TD	VA
Saleem et al. (2022)	ML	ANN, SVM	TD	CTC
Majumdar et al. (2021)	ML	LSTM	VS, TF, T, L, H	CTC
Jenifer and Priyadarsini (2022)	ML	MOLGBM	D, CV, instance ID, junction ID	TP
Yang et al. (2021)	DL	YOLO, LK	VS, TS, T	CTC
Zheng and Huang (2020)	DL	LSTM	TF, T	FTF
Singh et al. (2023)	DL	RNN, LSTM	TD, nodes (Junctions) and segments (Roads)	CTC
Anjaneyulu and Kubendiran (2022)	DL	CNN, SVM	T, TD	CTC
Qi and Cheng (2023)	DL	DSGCN	T, traffic network grids	CTC
Mohanty et al. (2020)	DL	LSTM	Region's travel, T, state of congestion, CV	VA
Cheng et al. (2021)	DL	CNN, LSTM, VAR	VS, TV, TS	Multi-feature prediction
Utomo et al. (2020)	DL	CNN	TD, CV	VA
Abdullah et al. (2023)	DL	BRNN, GRU	Number of lanes, roads conditions, accidents, CV, TD, traffic rate, traffic flow, S, weather, TF	CTC
Li et al. (2020)	ML, DL	LSTM-SPRVM	TF, S, road occupancy, rate, and TD	CTC
Mihaita et al. (2020)	DL	CNN, LSTM	TF, S, and occupancy	AD

Technique		Model		Parameters		Focus	
ML	Machine Learning	DBSCAN	Density-Based Spatial Clustering of Applications with Noise	VD	Vehicle Data	CTC	Control Traffic Congestion
DL	Deep Learning	CNN	Convolutional Neural Networks	RTL	Road and Traffic Length	ED	Emergency Detection
		BLSTME	Boosted Long Short-Term Memory Ensemble	AL	Accident Locations	VIA	Vehicle Illegal Activities
		AdaBoost	Adaptive Boosting	TV	Traffic Volume	VA	Vehicle Accumulation
		DBN	Deep Belief Network	TD	Traffic Density	FTF	Forecast Traffic Flow
		YOLO	You Only Look Once	VS	Vehicle Speed	TP	Traffic Prediction
		ANN	Artificial Neural Networks	TF	Traffic Flow	AD	Anomaly Detection
		SVM	Support Vector Machine	T	Time		
		LSTM	Long Short-Term Memory	L	Length		
		MOLGBM	Mayfly Optimization Light Gradient Boosting Machine	H	Headway		
		LK	Lucas and Kanade algorithm	D	Date		
		RNN	Recurrent Neural Network	CV	Count of Vehicle		
		DSGCN	Defect detection of Shield tunnel lining based on Graph Convolutional Networks	TS	Traffic State		
		VAR	Vector Auto-Regression Model				
		GRU	Gated Recurrent Unit				
		SPRVM	Spark Parallelization Relevance Vector Machine				
		BRNN	Bidirectional Recurrent Neural Networks				

such as only considering the traffic congestion prediction for a specific road in Canada, which may limit the generalizability of the proposed framework to other locations. Finally, the proposed research is theoretical and does not include any practical implementation or validation, which may limit the practical impact of the proposed framework.

In Mihaita et al. (2020), the research work presents a deep learning framework for predicting motorway traffic flow in Sydney, Australia. The approach involves data profiling, outlier identification, spatial and temporal feature generation, and deep learning model development. The study found that LSTM had the best predictive performance among the deep learning models tested and that the optimal past time horizon needs to be adapted for each deep learning model. Anomaly detection and outlier removal were shown to significantly improve prediction results and the amount of historical information needed to train and adjust the deep learning models. The strength of this work is the combination of data profiling, outlier detection, spatial and temporal feature generation, and multiple deep learning models to present an advanced deep learning framework for highway traffic flow prediction. On the other hand, the study is limited by the use of only one dataset, and future work will include designing traffic flow-based detection methods for larger transport networks and incorporating spatial relations between traffic stations using CNNs with graph-structured data. Another limitation of this study is that it only focuses on a single motorway in Sydney, Australia, so the findings may not be generalizable to other motorways or cities with different traffic patterns. Additionally, the study does not take into account external factors such as weather conditions or special events, which can significantly impact traffic flow and prediction accuracy. The study also only considers historical traffic flow data, without incorporating real-time data or other types of data such as road network information, which could potentially improve prediction accuracy.

Table 3
Evaluation of traffic management research prototypes considering the traffic management layer.

Ref	Traffic Management Layer		
	Decision Making	Performance Evaluation	Technology
Balasubramanian et al. (2023)	Modifies the timing of traffic signals	Simulations, Confusion matrix	IoT
Lilhore et al. (2022)	Updates traffic signal schedules	MATLAB simulation	IoT
Kothai et al. (2021)	Prediction of traffic congestion for traffic management	Python Tensor Flow API, SUMO and OMNeT++ 98% of accuracy, 96% of precision, and 94% of recall	IoT
Abbas (2019)	Vehicle illegal activities detection rate	OpenCV and Python tools. Detection rate: V-ITS: 90% CNN: 84%	-
Ravi et al. (2021)	Suggests alternative routes	Anaconda platform	YOLO
Abhijit et al. (2021)	Overcome the flaws of static timers	Python 3.5, along with TensorFlow and Keras	YOLOv3 and YOLOv5
Saleem et al. (2022)	Providing an alternative path for traveling	Simulation Accuracy 95%, miss rate 5%	IoT, MA
Majumdar et al. (2021)	Congestion prediction for smart sustainable cities	Python 3.7.1. Keras and SKLearn library 84–95% Accuracy	IoT
Jenifer and Priyadarsini (2022)	Traffic prediction for the smart city	Python 3.9, numpy and scikit learn libraries. accuracy 91.9%	IoT
Yang et al. (2021)	Congestion detection	Pytorch framework mAP 0.5 89.7%, FPS 44, Precision 89.2%, Recall 90.1%	YOLOv3
Zheng and Huang (2020)	Traffic flow management	MATLAB MAE 43.7798, RMSE 61.1699, MAPE 20.97%	-
Singh et al. (2023)	Traffic congestion management	Simulation	-
Anjaneyulu and Kubendiran (2022)	Short-term traffic congestion prediction	Python 3.10.5, API-Keras. Confusion matrix, Accuracy: 97.16% Precision: 98.21, recall: 98.8 F1-Score: 98.5	-
Qi and Cheng (2023)	Traffic congestion prediction	Pytorch 1.9.0 MAE 9.98, RMSE 16.63, MAPE 9.35%	-
Mohanty et al. (2020)	enhancing congestion management on neighborhood-wide scales	Simulation RMSE Gave the lowest value in different scenarios	-
Cheng et al. (2021)	Short-term traffic flow prediction	Simulation MSE 0.257 MAE 0.396 MAPE 0.507	-
Utomo et al. (2020)	Traffic congestion detection	Simulation Precisions of recording video UAV 90.75%, streaming video 90%	YOLO, UAV
Abdullah et al. (2023)	Optimizing traffic flow in smart cities and congestion prediction	OMNeT-4.6, SUMO-0.21.0, and INETMANET-2.0 framework.	-
Li et al. (2020)	Traffic congestion prediction and visualization	DataX and DataV tools, MF-TCPV framework Accuracy 92.36%	-
Mihaita et al. (2020)	Traffic congestion anomaly detection and prediction	Prototype MAPE = 13.68, MAE=143.61, RMSE=196.13	-

Technology	
IoT	Internet of Things
YOLO	You Only Look Once
MA	Mobile Applications
UAV	Unmanned Aerial Vehicle

5.3. Evaluation of major traffic management research

Our study of traffic management research includes a thorough examination of 20 research prototypes representing the most recent advances in the area. These prototypes were all published within the last 5 years, demonstrating the findings' relevance and timeliness. This evaluation guarantees that the insights gathered from the analysis are based on the most recent research and reflect current trends and advancements in traffic management by focusing on recent publications. These research prototypes serve as a solid platform for measuring development and innovation in the area, providing vital insights into the evolving tactics, technologies, and techniques that will shape the future of traffic management.

Tables 1, 2, and 3 summarize the main methodologies and techniques employed in traffic management and analysis. Several studies focus on real-time data capturing, such as collecting photos and vehicle location information using cameras and sensors. One notable option is an IoT-based ITM system that collects real-time and static data via a network of sensors. By altering the timing of traffic lights, Machine Learning (ML) techniques, notably Density-based spatial clustering of Applications with Noise (DBSCAN) clustering, are used to regulate traffic congestion. The obtained data is then processed and analysed in layers, using ML algorithms such as DBSCAN clustering and deep learning models such as CNN and LSTM. The analysis is to solve traffic-related concerns such as volume, congestion, and density. Traffic management decision-making entails measures such as adjusting traffic signal timing, offering alternate routes, and reducing road accidents. Simulations, confusion matrices, and accuracy measurements are used to evaluate performance.

To evaluate the prototype's effectiveness, simulations and performance evaluations utilizing confusion matrices are performed. Other research looks into the usage of deep learning models such as YOLO and CNN for traffic analysis. These models are trained using datasets comprising photos and labelled vehicle instances. Each prototype focuses on tasks such as traffic flow, density, and vehicle prediction. Comparisons of several model versions indicate increases in accuracy and speed, proving the effectiveness of these models in traffic control.

Furthermore, the assessed prototypes emphasize the significance of taking geographical and temporal elements into account in traffic analyses. To study traffic patterns, forecast traffic flow, and predict congestion, techniques such as spatial–temporal analysis, temporal analysis, and time-series analysis are used. The combination of IoT technology, ML algorithms, and simulation systems enables the creation of smart traffic management techniques that optimize traffic flow, reduce congestion, and improve road safety.

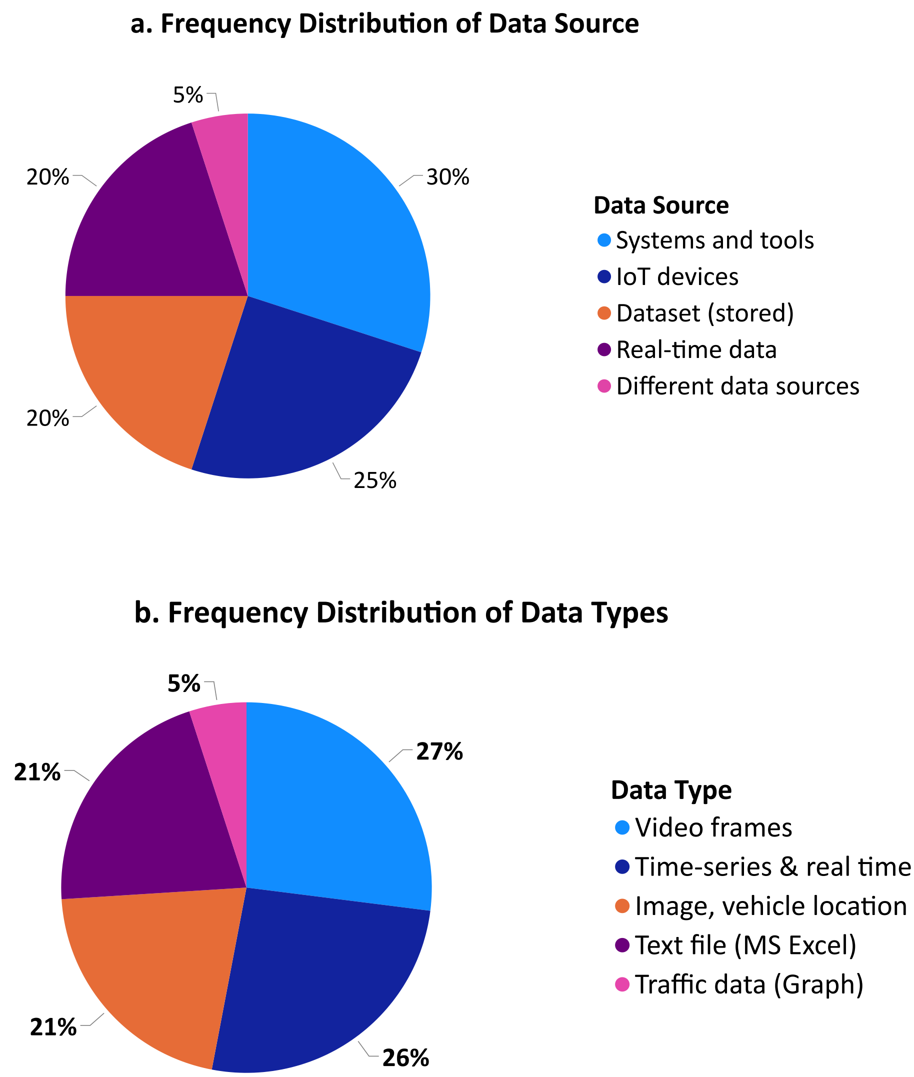


Fig. 4. The distribution of preferences across eight unique sections.

To solve diverse traffic-related issues, the surveyed research prototypes apply a variety of data collection methodologies, analysis layers, and traffic management approaches. The use of real-time data, ML algorithms, DL models, and IoT technology demonstrates the promise of advanced Traffic Management Systems (TMSs) in smart cities, providing increased decision-making skills, congestion reduction, and improved traffic flow.

Fig. 4 depicts the distribution of preferences across eight unique sections labelled "a" to "h", which represent the criteria used to evaluate traffic management layers. First, in terms of data sources, systems, and tools, the Data Acquisition Layer was the most popular, scoring 30% of the research prototypes, followed by IoT devices (25%), real-time data (20%), and saved dataset (20%). Video frames and time series along with real-time data were the most popular data categories, reaching 25% of the total, while image and vehicle position data and text files (MS Excel) were both close behind at 20%. When it came to data dimensions, temporal data was the dominator, recording 40% of the total, followed by geographic data (20%) and the integration of spatial and temporal data (20%). The majority of research prototypes (30%) focused on metropolitan roads and motorways, whereas roads in smart cities scored 20%.

Second, when looking at the specific focal areas, traffic congestion was the leading topic, reaching 40% of the research prototypes, followed by road accidents, road density, vehicle traffic intersections, and vehicle unlawful activities, each scoring 10%. Traffic prediction and high dimension reduction were less common, accounting for only 5% of the trials. Among the models utilized, LSTM was the most common (35%), followed by CNN (15%). These models produce remarkably precise outcomes.

Finally, in the context of decision-making, traffic congestion prediction received the most attention, reaching around 45%, followed by traffic flow management (30%), traffic signal timing management (15%), and traffic congestion detection (10%). In terms of technology, IoT emerged as the most widely used, reaching 40%, followed by YOLO (20%).

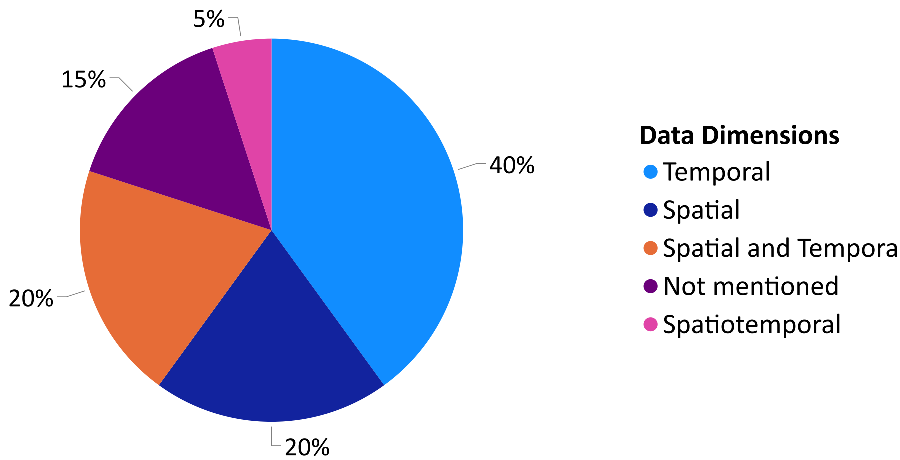
These findings highlight the importance of real-time data, temporal analysis, and LSTM models in traffic management and directing decision-making processes, such as traffic signal timing management and congestion prediction, as well as addressing other field priority areas.

6. Traffic management open issues and research direction

Traffic management is a complex problem that can benefit from the use of Deep Learning (DL) techniques. However, several issues need to be addressed when applying DL to this field. Some of these challenges and future research directions include:

1. Poor Traffic Congestion Prediction: With traffic congestion being part of the focus in the evaluated research prototypes (with

c. Frequency Distribution of Data Dimensions



d. Frequency Distribution of Type of Area

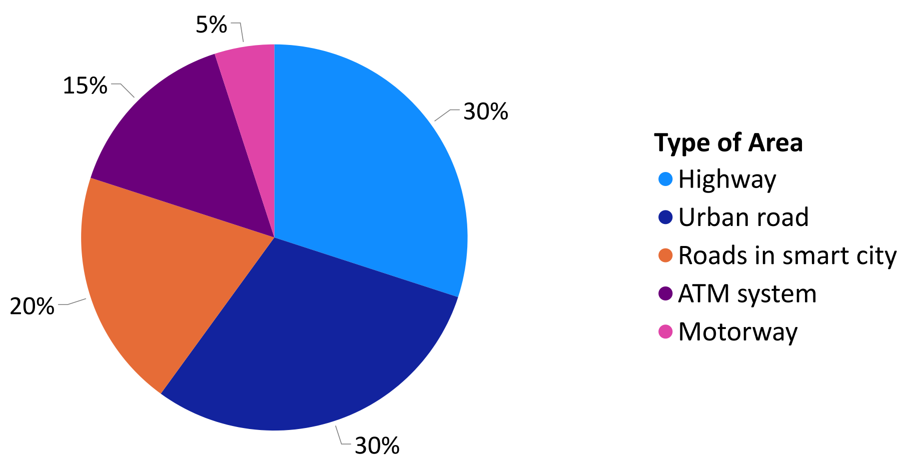


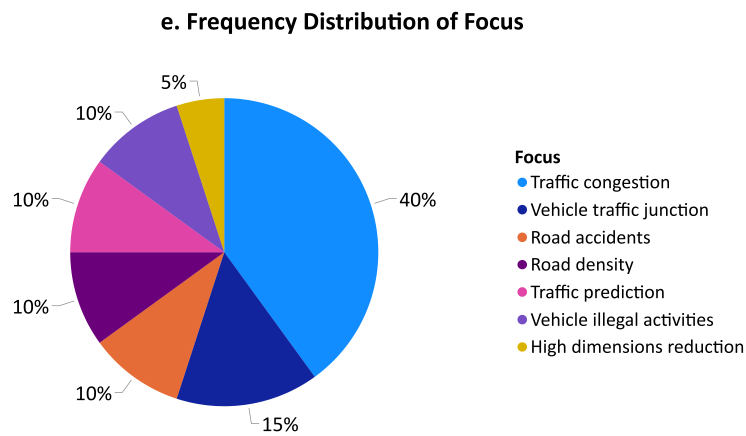
Fig. 4. (*continued*).

only 40% of the overall attention dedicated to it), the findings reveal the need for effective measures to alleviate congestion. Possible solutions could include implementing Intelligent Traffic Management (ITM) systems that utilize real-time data from Internet of Things (IoT) devices (i.e., only 25% of the evaluated research prototypes have used such technology that supports real-time data). It is worth noting that 30% of the evaluated research prototypes have used published datasets provided by systems and tools such as Kaggle and GRAM (Abbas, 2019), whereas not all published datasets represent real-time data which includes different weather scenarios and emergency situations. Furthermore, applying predictive models such as Long Short-Term Memory (LSTM) is a possible solution, while not many of the evaluated research prototypes have used such deep learning technique (e.g., only 35% have used LSTM for traffic congestion prediction) which can assist in proactive traffic management and rerouting strategies.

2. Lack of Traffic Flow Management: Given that traffic flow management has received some attention, however, the allocation of focus to traffic flow management remains restricted (i.e., only 30% focused on traffic flow management). Measures to increase traffic flow research prototypes can be examined. Implementing dynamic traffic signal timing management (approximately 15% of the evaluated research prototypes focused on this method) based on real-time data analysis and predictive algorithms could be one solution. This can aid in the optimization of signal timings to alleviate congestion and improve traffic flow at crossings. Future research could look into data fusion methodologies, data quality assurance, and privacy issues related to real-time data integration.

3. Technology Integration Challenges: Because IoT has emerged as the most widely used technology (only 40% of total usage), utilizing IoT devices to capture real-time traffic data will improve monitoring and control. Furthermore, the employment of computer vision techniques such as YOLO (which was used in just 20% of the evaluated research prototypes) can aid in traffic monitoring, vehicle tracking, and the identification of traffic offences, Which contributes to the traffic management field. Besides that, unlike previous studies that utilized YOLO v3 and v5 (Abhijit et al., 2021; Yang et al., 2021), it is noteworthy to consider the potential advantages of adopting the latest version, YOLO v8. Future studies could concentrate on incorporating developing technologies into TMSs, such as artificial intelligence, machine learning, and big data analytics. Exploring the possibilities of technologies such as blockchain, cloud computing, and self-driving cars can help to improve the efficiency, efficacy, and privacy of traffic management.

4. Decision Support Systems Leveraging: Given the emphasis on traffic congestion prediction (45%, which means that nearly half of



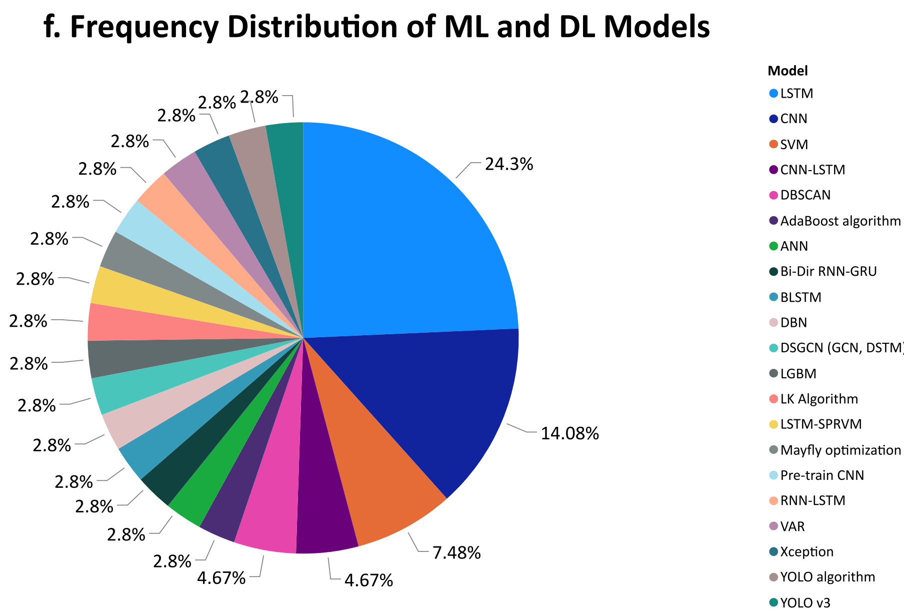


Fig. 4. (*continued*).

the studies examined in the evaluation) and 35% of the approaches, which means that roughly one-third of the evaluated studies used LSTM models as their chosen methodology for analysing and predicting traffic congestion, this highlights the popularity and effectiveness of LSTM models in this specific research area. Making educated traffic management decisions can be aided by establishing decision support systems that use these prediction models. These technologies can issue real-time alerts and recommend the best traffic control tactics. It is feasible to improve traffic management, reduce congestion, optimize traffic flow, and increase overall efficiency in transportation systems by addressing these challenges and adopting the suggested solutions based on the findings of the analysis.

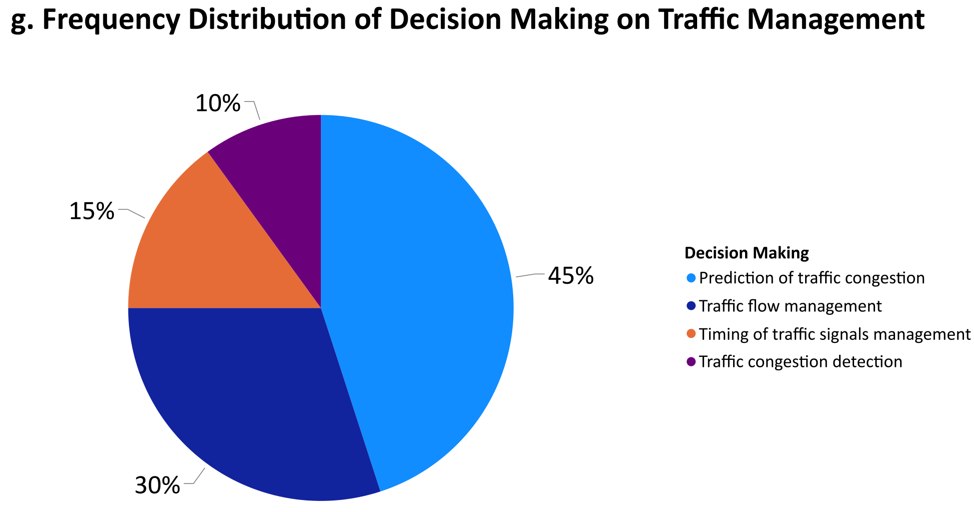
Furthermore, there are additional traffic management concerns and potential solutions that the results of the surveyed research may not have addressed. These include things like social media, weather data, holidays, driver behaviour, and social events. Future research could concentrate on the following areas to overcome these issues:

5. Insufficient Attention to Public Transportation Optimization: Current studies prioritizes the investigation of congestion-related issues over the assessment of public transportation systems. The optimization of public transport systems is one area that may have received insufficient attention (He et al., 2021). To solve this, the research could be done to discover solutions for increasing the efficiency of public transportation, such as introducing real-time tracking systems, optimizing bus routes, and integrating smart ticketing systems.

6. Overlooking the Potential of Sustainable Transit: The emphasis on traffic congestion and flow control may have overlooked the importance of encouraging sustainable transportation (de la Torre et al., 2021). Future research should look into measures to promote alternative means of transport, such as cycling, walking, and public transport, to minimize dependency on private vehicles and lessen traffic congestion and environmental impacts. Incorporating Human Activity Recognition (HAR) technology in pedestrian studies can provide a broader perspective on traffic congestion management by not only analysing vehicular traffic but also understanding and accounting for pedestrian movements and behaviours in urban settings (He et al., 2021; de la Torre et al., 2021).

7. Driver Behaviour Analysis: Understanding driver behaviour and promoting safe driving practices could make a significant contribution to traffic control efforts (ElSahly and Abdelfatah, 2022). To address concerns such as aggressive driving, distracted driving, and adherence to traffic rules and regulations, studies concentrating on driver behaviour analysis, education, and awareness campaigns could be done.

8. Emergency Management and Incident Response: The analysis may not have included emergency management and incident response



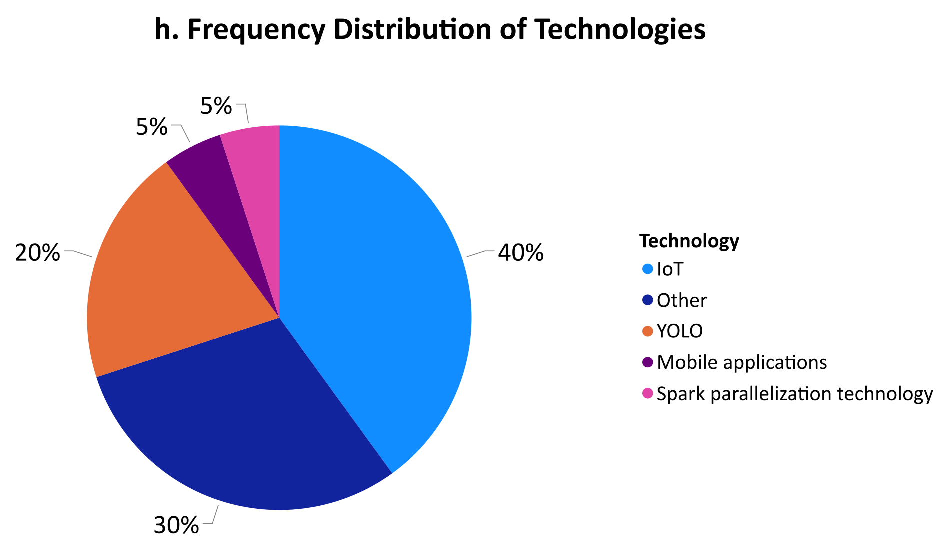


Fig. 4. (*continued*).

in traffic management sufficiently (Nama et al., 2021). More research might be conducted to investigate solutions for effective incident detection, rapid response, and traffic redirection during accidents, or road incidents to minimize disruptions and ensure efficient traffic flow.

9. Uncovering the Potential of Multi-City and Cross-Region Studies: Conducting research across various cities or regions can provide a more comprehensive view of traffic management. Comparing and assessing various urban environments, traffic patterns, and infrastructure can lead to complete and more adaptive solutions. Also, there were a few studies proposed working on different scenarios (Balasubramanian et al., 2023; Lilhore et al., 2022). The limited usage of different scenarios, on the other hand, obstructs the study of a wide range of potential obstacles and alternative tactics, thus omitting significant elements that may influence the effectiveness of congestion management efforts in the real world. Subsequent investigations may broaden by employing varied comparison evaluations among various areas to uncover subtle patterns. Gaining more insight into multi-city dynamics may be possible through longitudinal research that monitors temporal trends and uses cutting-edge analytical methods like machine learning or spatial modelling. Comprehensive understandings could be obtained by collaborative interdisciplinary techniques integrating sociology, economics, and urban planning.

10. Traffic Policy and Governance Considerations: Examining traffic management policy and governance frameworks can provide insights into regulatory impediments, legal ramifications, and the roles of various stakeholders (de la Torre et al., 2021). Future research could concentrate on identifying policy gaps, formulating policy recommendations, and investigating the institutional dynamics involved in implementing effective traffic management measures.

Comprehensive strategies for traffic management can be developed by taking into account these additional issues and future research directions, as well as exploring potential solutions. These strategies can include aspects such as public transportation optimization, sustainability, driver behaviour, and emergency management. These areas of emphasis have the potential to enhance the overall efficacy and efficiency of TMSs.

7. Conclusion

Traffic management has developed into a popular research area in recent years, attracting the interest of several researchers to tackle issues including flow prediction and traffic congestion. In this article, we thoroughly evaluated and analysed a wide range of strategies, methodologies, and technologies used in traffic management and discussed future research directions. Moreover, we present a generic traffic management architecture that has three different layers including the data acquisition layer, the data analysis layer, and the traffic management layer. In each layer, we identify a set of assessment criteria to evaluate 20 recently proposed research prototypes (i.e., published since 2019). Furthermore, we reveal 10 traffic management open issues and provide potential direction to overcome them. The traffic management open issues include poor traffic congestion prediction, lack of traffic flow management, technology integration challenges, decision support systems leveraging, insufficient attention to public transportation optimization, overlooking the potential of sustainable transit, driver behaviour analysis, emergency management and incident response, uncovering the potential of multi-city and cross-region studies, and traffic policy and governance considerations.

Throughout the work, it became clear that combining machine learning, deep learning, and the IoT has the potential to transform traditional traffic management practices. These cutting-edge technologies provide exciting potential for real-time data collection, analysis, and decision-making, resulting in more efficient and effective congestion mitigation strategies. This survey article provides insights for researchers, policymakers, and practitioners for designing and implementing successful traffic management methods by leveraging machine learning, deep learning, and IoT. Our survey is focused on traffic management, in contrast to other recent research surveys that have been published in the literature and that concentrate on target recognition using UAVs, traffic signal control, traffic congestion management, or traffic flow prediction. In this study, we provide a comprehensive overview of the advantages, disadvantages, and real-world applications of using machine learning and deep learning in traffic management. We leverage the power of machine learning and deep learning techniques to highlight outstanding concerns and future directions in the field of traffic management, based on the evaluation results. Continued research and development in this field are critical to addressing the chronic issue of traffic congestion and developing sustainable and efficient transport systems for future urban environments.

CRediT authorship contribution statement

Hanan Almukhalfi: Data curation, Formal analysis, Investigation, Methodology, Resources, Validation, Visualization, Writing – original draft. **Ayman Noor:** Conceptualization, Data curation, Funding acquisition, Investigation, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Talal H. Noor:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data generated or analysed during this study are included in this published article.

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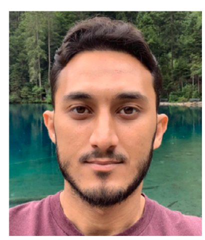

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